

TASK 3

Automation System Design

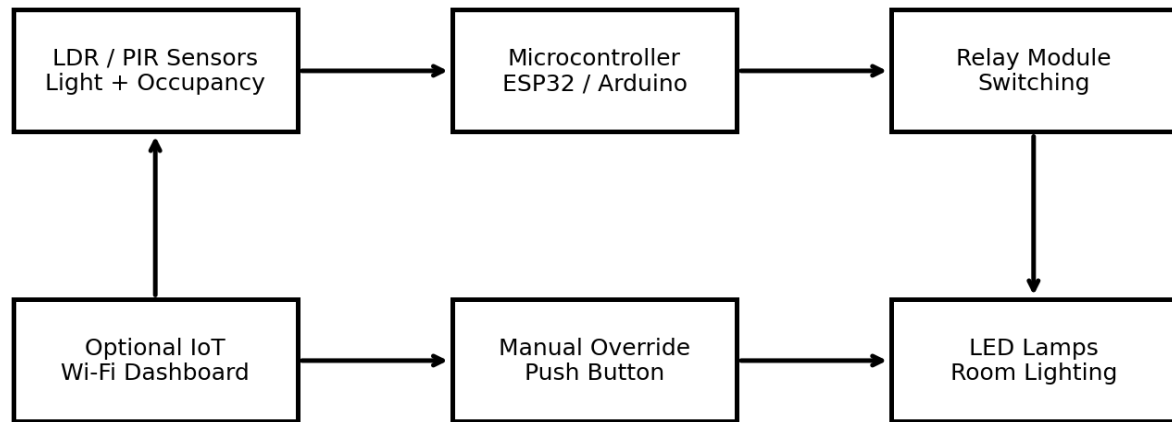
Smart Home Lighting Automation System

STUDENT	Onimago Ibrahim
DOMAIN	Robotics & Automation
SYSTEM	Smart Home Lighting
CONTROL CORE	ESP32 / Arduino-class Microcontroller
INPUTS	LDR + PIR + Manual Override
OUTPUT	Relay-driven Lamp

A practical sensor-based automation concept designed to reduce unnecessary lighting and provide automatic occupancy-aware control.

1. System Architecture

SMART HOME LIGHTING — SYSTEM ARCHITECTURE

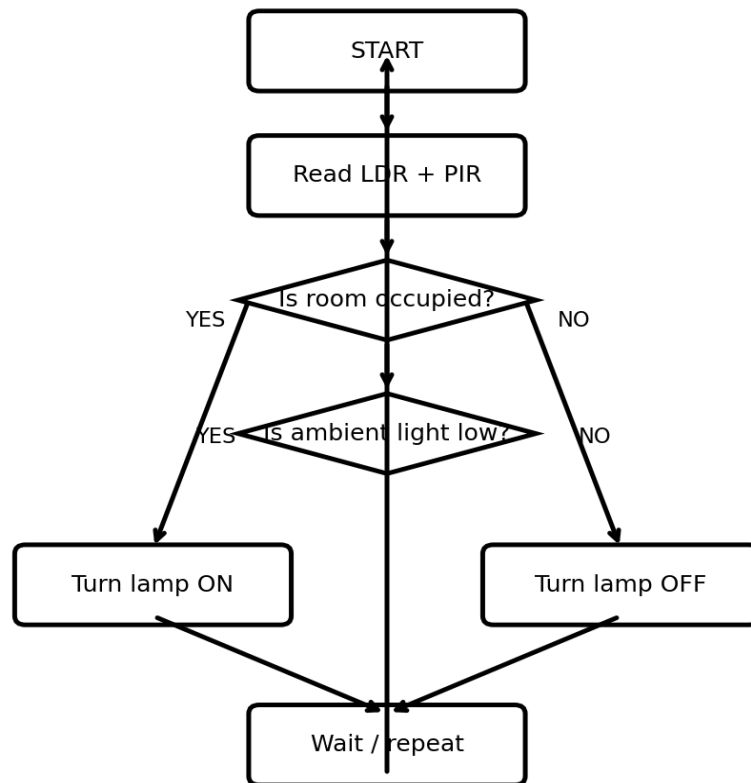


Rule: occupancy detected + ambient light below threshold → lamp ON; otherwise OFF.

The system uses an LDR to estimate ambient light and a PIR sensor to detect occupancy. The microcontroller evaluates these inputs and controls a relay module connected to the lamp. A push button provides manual override. An optional Wi-Fi dashboard can be added for monitoring and remote control.

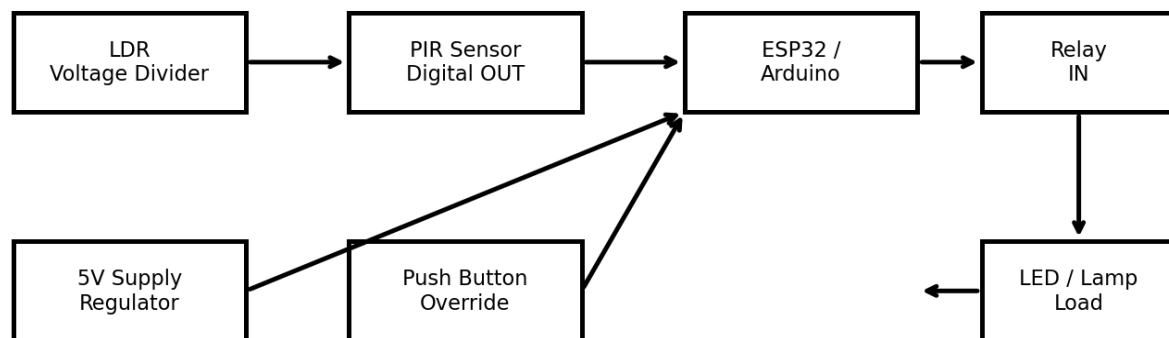
2. Operating Logic

Smart Home Lighting Control Flowchart



3. Circuit / Wiring Concept

SMART HOME LIGHTING — CIRCUIT / WIRING CONCEPT



For real AC installation, use an appropriately rated isolated switching device and qualified electrical practices.

4. Components and Functions

- LDR: measures ambient light level.
- PIR sensor: detects human movement/occupancy.
- ESP32/Arduino: reads sensors and executes the control logic.
- Relay module: electrically switches the lighting load.
- Push button: provides manual override.
- LED/lamp: controlled output.

5. Example Control Rule

If the room is occupied and the ambient light is below the selected threshold, the controller activates the lamp. If the room is unoccupied or sufficient natural light is present, the controller turns the lamp off. A practical implementation should add debounce/filtering and a short delay to avoid rapid switching.

6. Testing Plan

- Test 1: Cover the LDR while occupancy is detected → lamp should switch ON.
Test 2: Increase ambient light above the threshold → lamp should switch OFF.
Test 3: Remove occupancy → lamp should switch OFF after the configured delay.
Test 4: Press manual override → lamp state should follow the override rule.

Safety note: the diagram is a low-voltage control concept. Any mains-powered lighting connection requires correctly rated isolation/switching hardware and appropriate electrical safety practices.